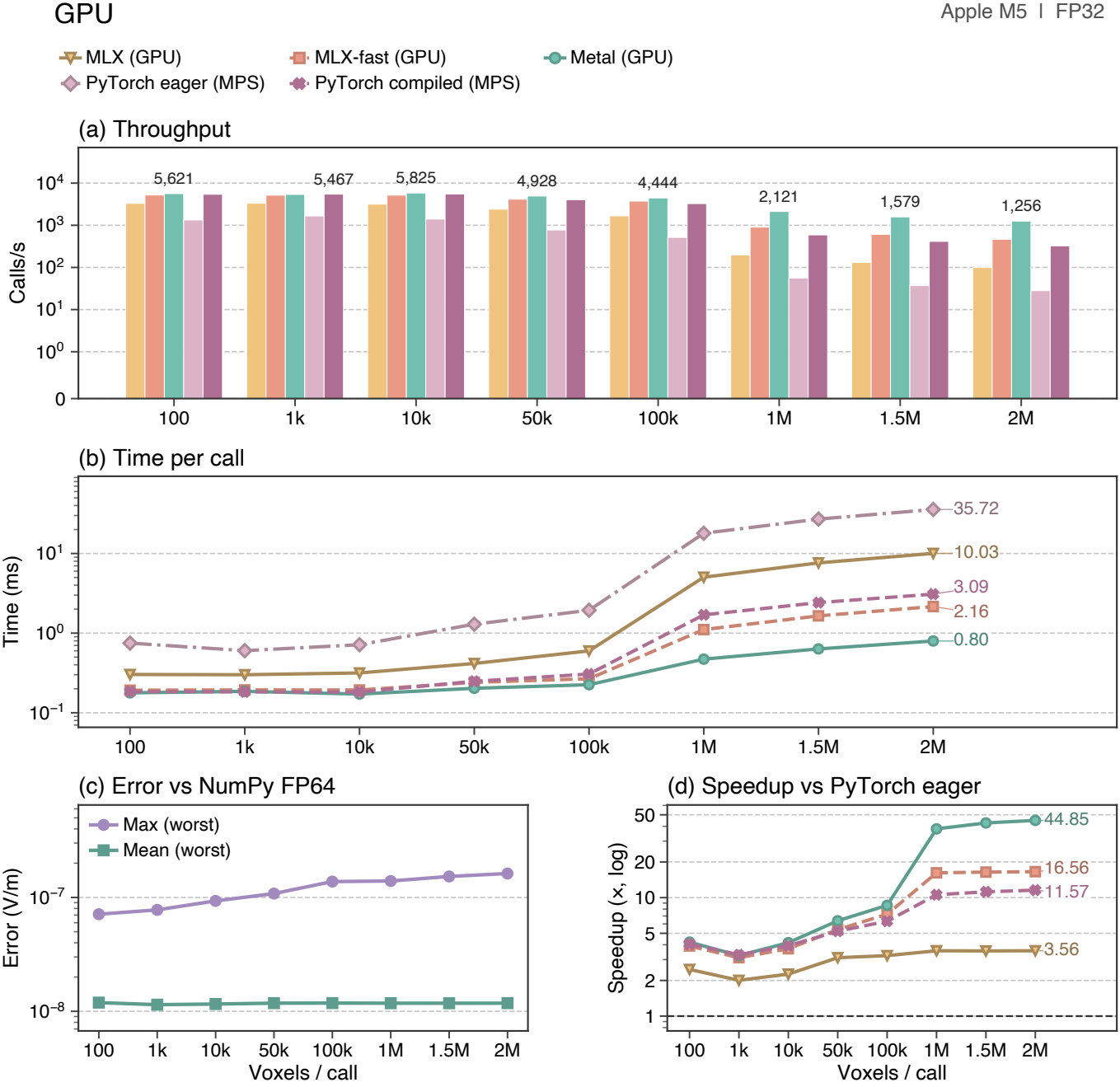
GPU
Apple M5 | FP32
MLX (GPU)
MLX-fast (GPU)
Metal (GPU)
PyTorch eager (MPS)
PyTorch compiled (MPS)
(a) Throughput
Calls/s
5,621
5,467
5,825
4,928
4,444
2,121
1,579
1,256
100
1k
10k
50k
100k
1M
1.5M
2M
(b) Time per call
Time (ms)
35.72
10.03
3.09
2.16
0.80
(c) Error vs NumPy FP64
Max (worst)
Mean (worst)
Error (V/m)
Voxels / call
(d) Speedup vs PyTorch eager
Speedup (×, log)
44.85
16.56
11.57
3.56
Voxels / call